ALGEBRAIC REPRESENTATION AND MANIPULATION

Past Paper Topical Questions (Paper 4)

1. Simplify. Q(3b) 0580/41/M/J/20

(i) $3a-5b-a+2b$

..... [2]

(ii) $\frac{5}{3x}\times\frac{9x}{20}$

..... [2]

2. Expand and simplify. Q(3e) 0580/41/M/J/20

$(3x-5y)(2x+y)$

..... [2]

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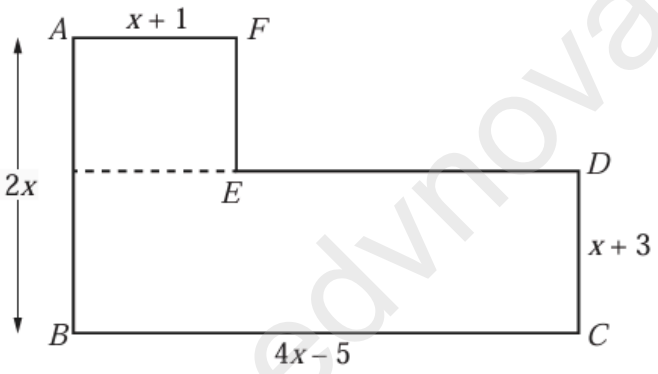
3. Solve the equation. $\frac{2x+5}{3-x} = \frac{14}{15}$

Q(4b) 0580/43/M/J/20

$x =$ ..... [3]

4. All the lengths in this question are in centimetres.

Q(5a) 0580/43/M/J/20



NOT TO SCALE

The diagram shows a shape $ABCDEF$ made from two rectangles.
The total area of the shape is $342\,\text{cm}^2$.

(a) Show that $x^2 + x - 72 = 0$.

[5]

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5. **(a)** Simplify. Q(7) 0580/41/M/J/21

$$\frac{x^2-25}{x^2-x-20}$$

..... [3]

(b) Write as a single fraction in its simplest form.

$$\frac{x+5}{x}+\frac{x+8}{x-1}$$

..... [3]

6. **(d)** Factorise the following completely. Q(3d) 0580/42/M/J/21

(i) $x^2-3x-28$

..... [2]

(ii) $7(a+2b)^2+4a(a+2b)$

..... [2]

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7. (a) (i) The equation $y = x^3 - 4x^2 + 4x$ can be written as $y = x(x-a)^2$. **Q(9a) 0580/42/M/J/21**

Find the value of a.

$a =$ ..... [2]

8. (a) $y = px^2 + t$ **Q(2) 0580/43/M/J/21**

(i) Find the value of y when $p = 3$, $x = 2$ and $t = -13$.

$y =$ ..... [2]

(ii) Rearrange the formula to write x in terms of p, t and y.

$x =$ ..... [3]

(b) (i) Factorise.

$15x^2 - 2x - 8$

..... [2]

(ii) Solve the equation.

$15x^2 - 2x - 8 = 0$

$x =$ ..... or $x =$ ..... [1]

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(c) Factorise completely.

$$x^3 - 16xy^2$$

..... [3]

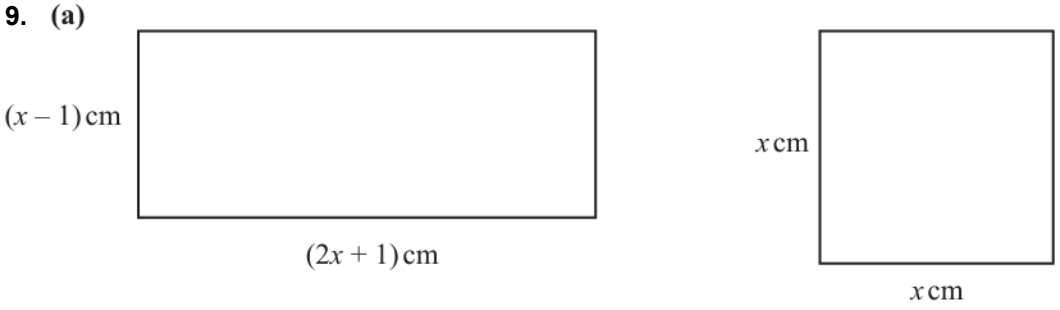
(d) Simplify.

$$\frac{2x - 1 - 4ax + 2a}{2x^2 - x}$$

..... [4]

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9. (a) Q(9) 0580/41/M/J/22



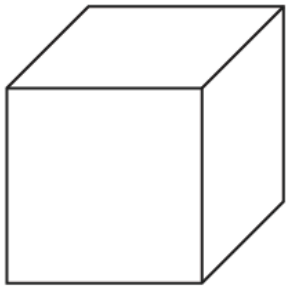
The area of the rectangle is $29\,cm^2$ greater than the area of the square.
The difference between the perimeters of the two shapes is k cm.

Find the value of k.
You must show all your working.

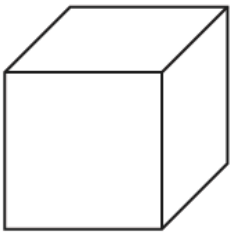
$k =$ ..... [6]

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(b)



$(y + 1)$ cm



y cm

NOT TO SCALE

The volume of the larger cube is $5\,\text{cm}^3$ greater than the volume of the smaller cube.

(i) Show that $3y^2 + 3y - 4 = 0$.

[4]

(ii) Find the volume of the smaller cube.
Show all your working and give your answer correct to 2 decimal places.

..... cm^3 [4]

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10. **(a)** Solve. Q(8) 0580/42/M/J/22

$$10-3p=3+11p$$

$p =$ ..... [2]

(b) Make m the subject of the formula.

$$mc^2-2k=mg$$

$m =$ ..... [3]

(c) Solve.

$$\frac{1}{x-3}+\frac{4}{2x+3}=1$$

$x =$ ..... or $x =$ ..... [5]

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(d) Solve the simultaneous equations.
You must show all your working.

$$x+2y=12$$
$$5x+y^2=39$$

$x =$ ..... $y =$ .....

$x =$ ..... $y =$ ..... [5]

(e) Expand and simplify.

$$(2x-3)(x+6)(x-4)$$

..... [3]

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11. (a) Simplify. Q(6) 0580/43/M/J/22

$$a-2b-3a+7b$$

..... [2]

(b) Expand and simplify.

$$4(x-5)-(3-2x)$$

..... [2]

(c) Write as a single fraction in its simplest form.

$$\frac{3}{x-5}-\frac{7}{2x}$$

..... [3]

(d) Solve.

$$\frac{13-4x}{3}=6-x$$

$x =$ ..... [3]

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(e) Make x the subject of the formula.

$$y = \frac{5(p-2x)}{x}$$

$x =$ ..... [4]

12. (c) Expand and simplify. Q(5cde) 0580/42/M/J/24

$$(x+4)(x-3)(3x-1)$$

..... [3]

(d) (i) Show that $(3x+5)+\frac{7}{x-2}=x$ simplifies to $2x^2+x-3=0$.

[4]

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(ii) Solve by factorisation $2x^2 + x - 3 = 0$.

$x =$ ..... or $x =$ ..... [3]

(e) A solid cylinder has base radius x and height $3x$.
The **total** surface area of the cylinder is the same as the **total** surface area of a solid hemisphere of radius $5y$.

Show that $x^2 = \frac{75y^2}{8}$.

[The surface area, A, of a sphere with radius r is $A = 4\pi r^2$.]

[4]

13. Expand. **Q(6) 0580/41/M/J/25**

$5x^2(3x - 2)$

..... [2]

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14. **(a)** Write down **all** the factors of 18. Q(10) 0580/41/M/J/25

..... [2]

(b) Factorise.

$3y - xy + 15 - 5x$

..... [2]

(c) $3y - xy + 15 - 5x = 18$

where x and y are positive integers.

Using your answers to **part (a)** and **part (b)**, find one possible value of x and the corresponding value of y.

$x =$ ..... , $y =$ ..... [2]

15. Make x the subject of this formula. Q(13) 0580/41/M/J/25

$A = w^2 + 5x^2$

$x =$ ..... [3]

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16. $2x^2+12x-2$ can be written in the form $a(x+b)^2-c$. **Q(18) 0580/41/M/J/25**

Find the values of a, b and c.

$a =$ ..... , $b =$ ..... , $c =$ ..... [3]

17. $y = mx+c$ **Q(7) 0580/42/M/J/25**

Rearrange the formula to make m the subject.

$m =$ ..... [2]

18. Factorise. **Q(14) 0580/42/M/J/25**

$5x-10-ax+2a$

..... [2]

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19. Simplify. Q(23) 0580/42/M/J/25

$$\frac{h^2+4h}{h^2-16}$$

..... [3]

20. Factorise. Q(8) 0580/43/M/J/25

(a) $28xy-12x$

..... [2]

(b) $y-6x+2xy-3$

..... [2]

21. Expand and simplify. Q(10) 0580/43/M/J/25

(a) $2x-x(5-x^2)$

..... [2]

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(b) $(x+4)(x-5)(x+3)$

..... [3]

22. **(a)** Expand and simplify. Q(5) 0580/42/F/M/23

$$(2p^2-3)(3p^2-2)$$

..... [2]

(b) $s=\frac{1}{2}(u+v)t$

(i) Find the value of s when $u=20$, $v=30$ and $t=7$.

$s =$ ..... [2]

(ii) Rearrange the formula to write v in terms of s, u and t.

$v =$ ..... [3]

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(c) Factorise completely.

(i) $2qt-3t-6+4q$

..... [2]

(ii) x^3-25x

..... [3]

23. Simplify. Q(2) 0580/42/F/M/25

$$4y^2+3y-y^2+2y$$

..... [2]

24. **(a)** Factorise. Q(18) 0580/42/F/M/25

$$18a^2-98$$

..... [2]

(b) Expand and simplify.

$$(x+4)(2x-1)(x-2)$$

..... [3]

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25. $p^2 = r^2 - 8t$ Q(7) 0580/42/F/M/26

(a) Find p when $r = -15$ and $t = 7$.

$p =$ ..... [2]

(b) Rearrange $p^2 = r^2 - 8t$ to make t the subject.

$t =$ ..... [2]

26. (a) Factorise completely. Q(8) 0580/41/O/N/20

$3a^2b - ab^2$

..... [2]

(b) Solve the inequality.

$3x + 12 < 5x - 3$

..... [2]

(c) Simplify.

$(3x^2y^4)^3$

..... [2]

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(d) Solve.

$$\frac{2}{x} = \frac{6}{2-x}$$

$x =$ ..... [3]

(e) Expand and simplify.

$$(x-2)(x+5)(2x-1)$$

..... [3]

(f) Alan invests \$200 at a rate of r% per year compound interest.
After 2 years the value of his investment is \$206.46 .

(i) Show that $r^2 + 200r - 323 = 0$.

[3]

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(ii) Solve the equation $r^2 + 200r - 323 = 0$ to find the rate of interest.
Show all your working and give your answer correct to 2 decimal places.

$r =$ ..... [3]

27. **(a)** Find the integer values that satisfy the inequality $2 < 2x \leqslant 10$. Q(9) 0580/43/O/N/20

..... [2]

(b) Factorise completely.

(i) $6y^2 - 15xy$

..... [2]

(ii) $y^2 - 9x^2$

..... [2]

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(c) Simplify.

$$\frac{3}{x-1}-\frac{2}{2x+1}$$

..... [3]

(d) The straight line $y=3x+2$ intersects the curve $y=2x^2+7x-11$ at two points.

Find the coordinates of these two points.
Give your answers correct to 2 decimal places.

(..... , ...)

(..... , ...) [6]

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28. (a) $s = \frac{1}{2}at^2$ Q(2) 0580/41/O/N/23

Find the value of s when $a = 9.8$ and $t = 20$.

$s =$ ..... [2]

(b) Solve.

$5(4y-3) = 15$

$y =$ ..... [3]

(c) Expand and simplify.

$3(5x-8)-2(3x-7)$

..... [2]

(d) Rearrange $A = 2b^2 - 3c^3$ to make c the subject.

$c =$ ..... [3]

(e) Factorise completely.

$6pq-4q-3p+2$

..... [2]

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29. **(a)** Expand and simplify. Q(10ab) 0580/43/O/N/23

$$4(2x-1)-6(3-x)$$

..... [2]

(b) Factorise completely.

(i) $6x^2y+9xy$

..... [2]

(ii) $4x^2-y^2+8x+4y$

..... [3]

30. **(d)** Solve. Q(1de) 0580/41/O/N/24

$$\frac{34+2x}{5}=4-x$$

$x =$ ..... [3]

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(e) $P = d + \sqrt[3]{m^2}$

(i) Find P when $d = 7$ and $m = -8$.

$P =$ ..... [2]

(ii) Rearrange the formula to make m the subject.

$m =$ ..... [3]

31. (a) Simplify. Q(3) 0580/42/O/N/24

(i) $3m - 5n - 4m + 8n$

..... [2]

(ii) $(3a^2c^3)^4$

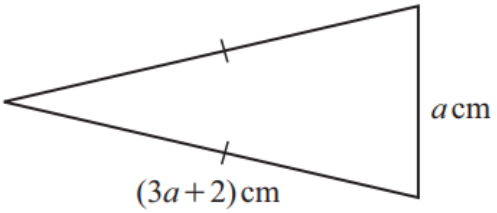
..... [2]

(iii) $\frac{4x}{5} - \frac{3x}{10} + \frac{2x}{15}$

..... [2]

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(b) This isosceles triangle has a perimeter of 35.5 cm.



NOT TO SCALE

Find the value of a.

$a =$ ..... [3]

(c) Using the quadratic formula, solve $5x^2 - 4x - 3 = 0$.
You must show all your working.

$x =$ ..... or $x =$ ..... [3]

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(d) Solve these simultaneous equations.

$$y = x^2 - 4x + 5$$
$$y = 2x - 3$$

You must show all your working.

$x =$ ..... $y =$ .....

$x =$ ..... $y =$ ..... [5]

32. (b) Expand and simplify $(x-1)(x+2)(x+3)$. **Q(6bc) 0580/43/O/N/24**

..... [3]

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(c) Solve the equation $2x^2 + x - 5 = 0$.
You must show all your working and give your answers correct to 2 decimal places.

$x =$ ..... or $x =$ ..... [4]

33. Expand. **Q(6) 0580/41/O/N/25**

$$g(3-2g)$$

..... [1]

34. Simplify. **Q(13) 0580/42/O/N/25**

$$\frac{7}{2m} + \frac{3}{8m}$$

..... [2]

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35. Expand and simplify. **Q(17) 0580/42/O/N/25**

$$(x-2)(2x+3)(x+4)$$

..... [3]

36. $G = \frac{4}{5}m^2n$ **Q(3) 0580/43/O/N/25**

Find the value of G when $m = 6$ and $n = 15$.

$G =$ ..... [1]

37. Expand and simplify. **Q(16) 0580/43/O/N/25**

(a) $7(x+2)+4(3x-5)$

..... [2]

(b) $(3x-y)(5x+2y)$

..... [2]

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38. Make t the subject of the formula. **Q(17) 0580/43/O/N/25**

$$x = \frac{7t}{5-t}$$

$t =$ ..... [3]

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